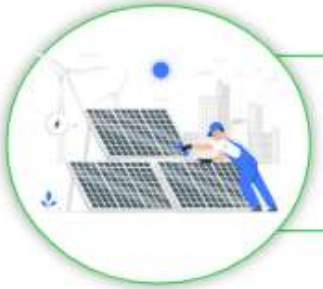
Module 9: Customer Orientation for a Solar PV System

Unit 9.1: Demonstrate the working principle of the solar PV system

Why is Customer Orientation Important?



Builds Confidence : Increases customer confidence in using the system.



Ensures Safety: Teaches safe handling.

Prolongs System Life : Encourages regular maintenance



Manage Expectations: Cleary communicates system performance under various conditions.



9.1.1: Demonstrate Start-up and Shutdown Procedures





Demonstrate Start-up and Shutdown Procedures







As a solar PV installer, it's important for you to familiarize your customer with the startup and shutdown procedures of the Solar PV system

Once commissioned, a PV system does not need to be shut down and started up under regular operation.



The customer needs to shut down and start up the system under specific circumstances



There is any malfunction that can be observed in the inverters or batteries such as unusual noise, overheating, fumes etc.



The customer finds voltage fluctuations in the electricity supply and is not sure of the reason.



There is a safety event related to the PV system or to the general electrical wiring in the facility.



The customer is instructed to switch off the system by the service provider.

Shut - Down Sequence



DCDB → Inverter → ACDB → Main Service Panel → Utility

DC Input Switch

AC Input Switch

Step Wise Process

Shut down sequence

- Step 1: Switch off the AC input or AC isolator switch on the solar inverter.

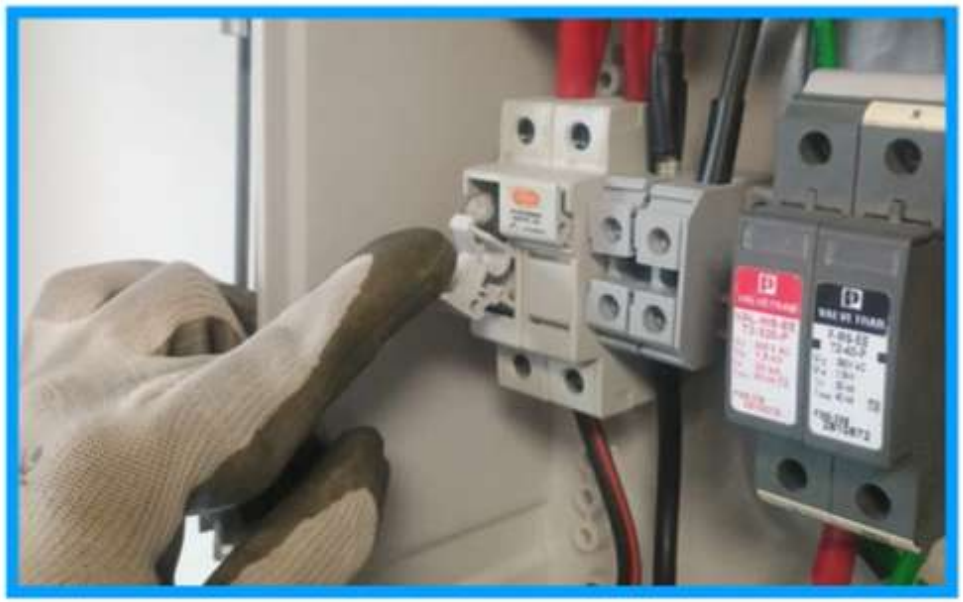




Step Wise Process

Shut down sequence

- Step 2: Switch off the DC input switch, which connects to the solar array.







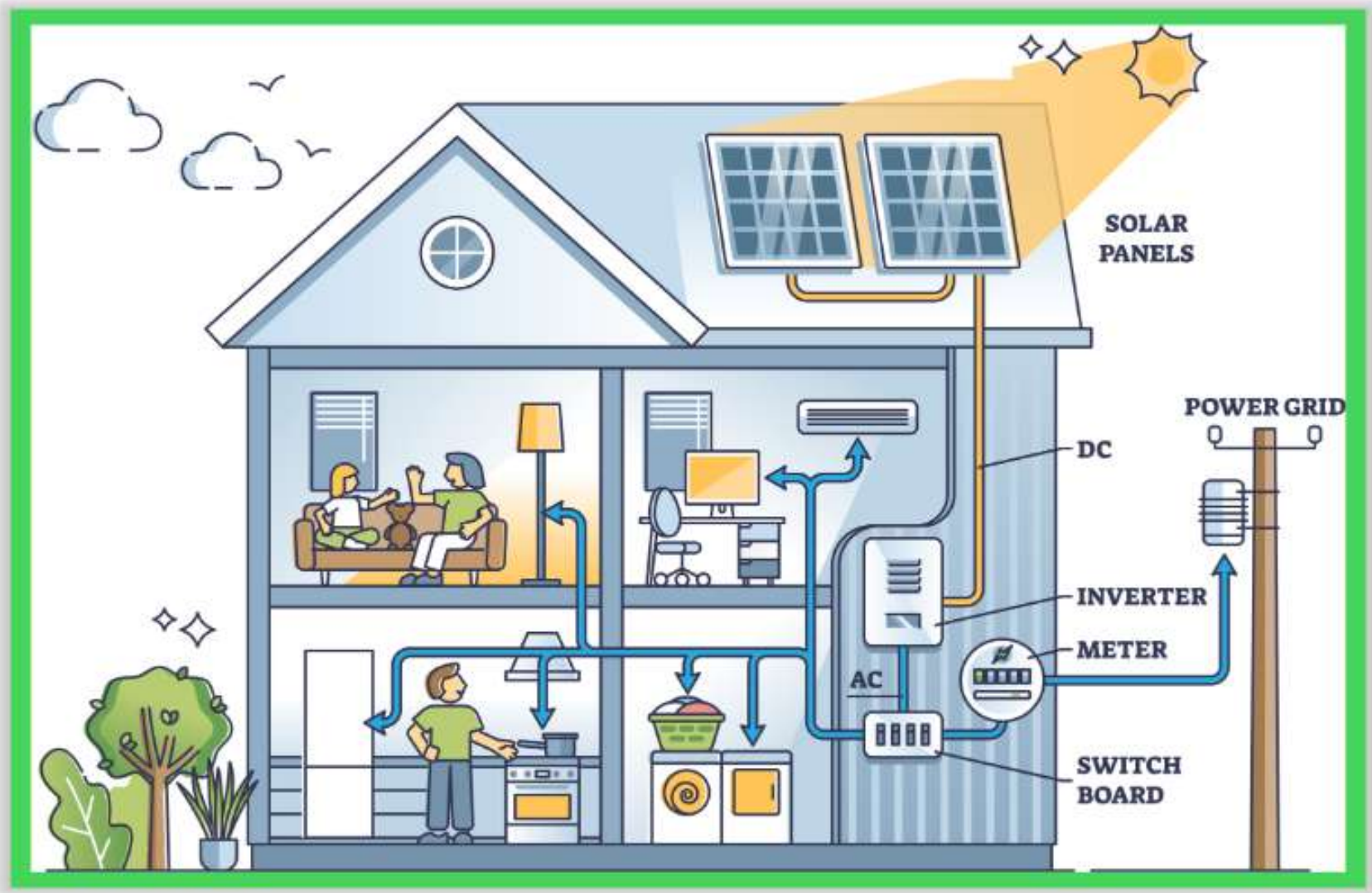
REMEMBER!

It is extremely important to follow the shutdown sequence because the probability of arcing is more at DC breaker than at AC breaker.

So, the AC breakers are switched off first to isolate load and reduce the generated current.

This is safer and leads to increased life of the protection devices used in the circuit.

The start-up procedure is just the reverse of the shut – down process.


SOLAR
PANELS
POWER GRID
DC
INVERTER
METER
AC
SWITCH
BOARD

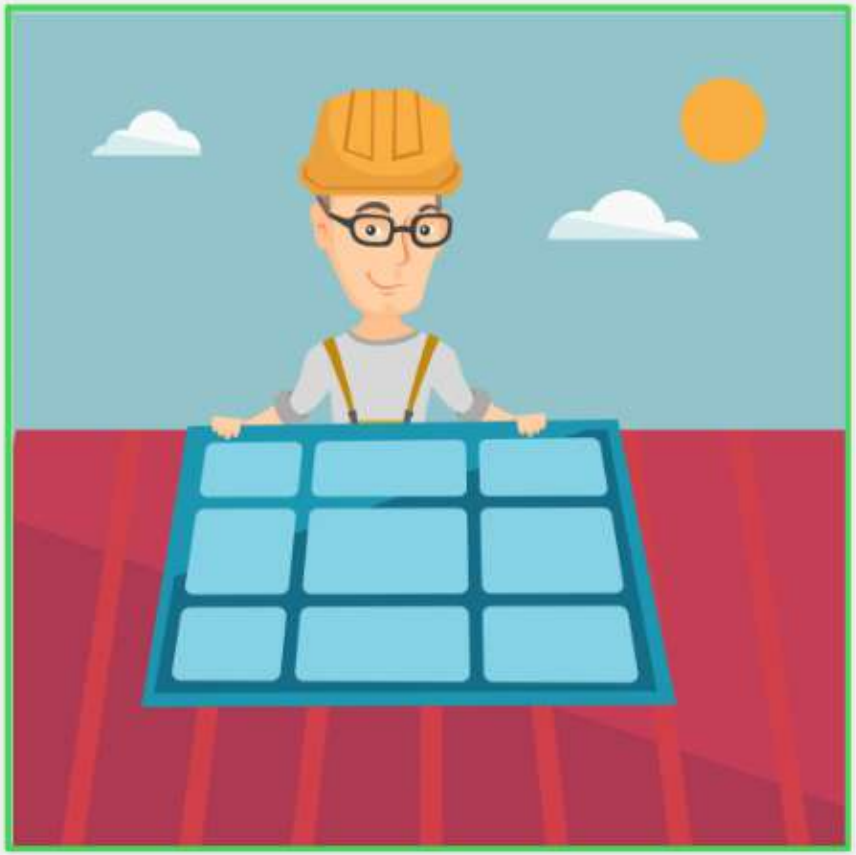


However, following basic procedures to clean the PV modules is essential for good performance of the system.



Clean the panels without stepping on them or placing heavy objects on them.

Proper System Maintenance Guidance provided by you to a customer will ensure its efficiency and longevity!



9.1.3:Explain What the Customer Should Expect from the System

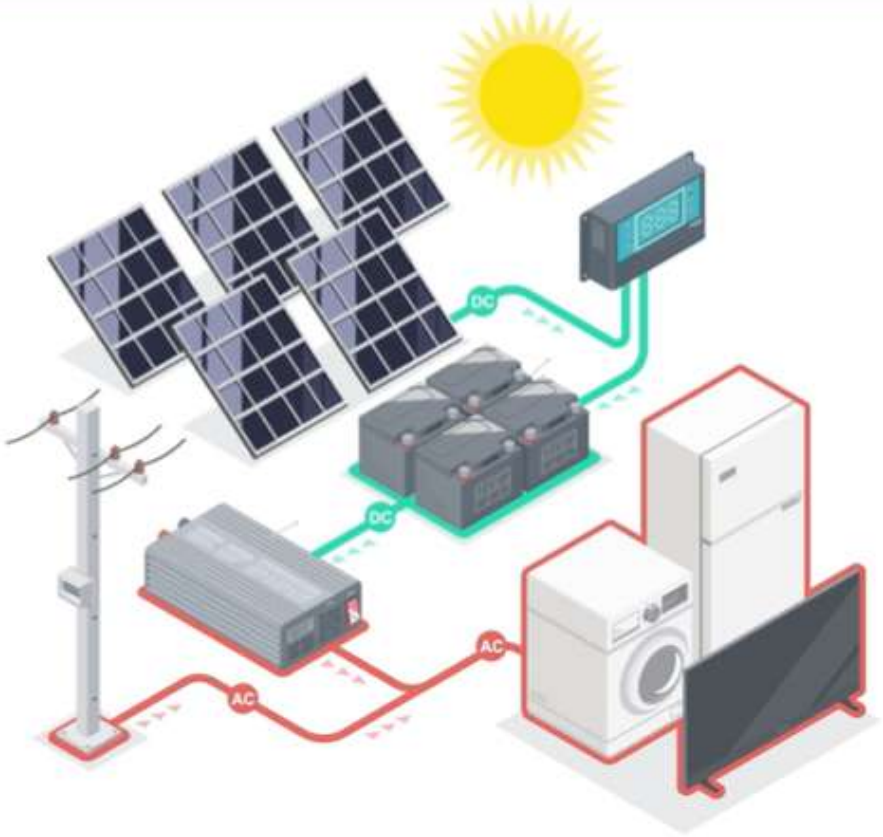


- There may be significant variation in how the Solar PV system performs under varying conditions.
- Be ready to provide answers to questions or doubts customers may have.

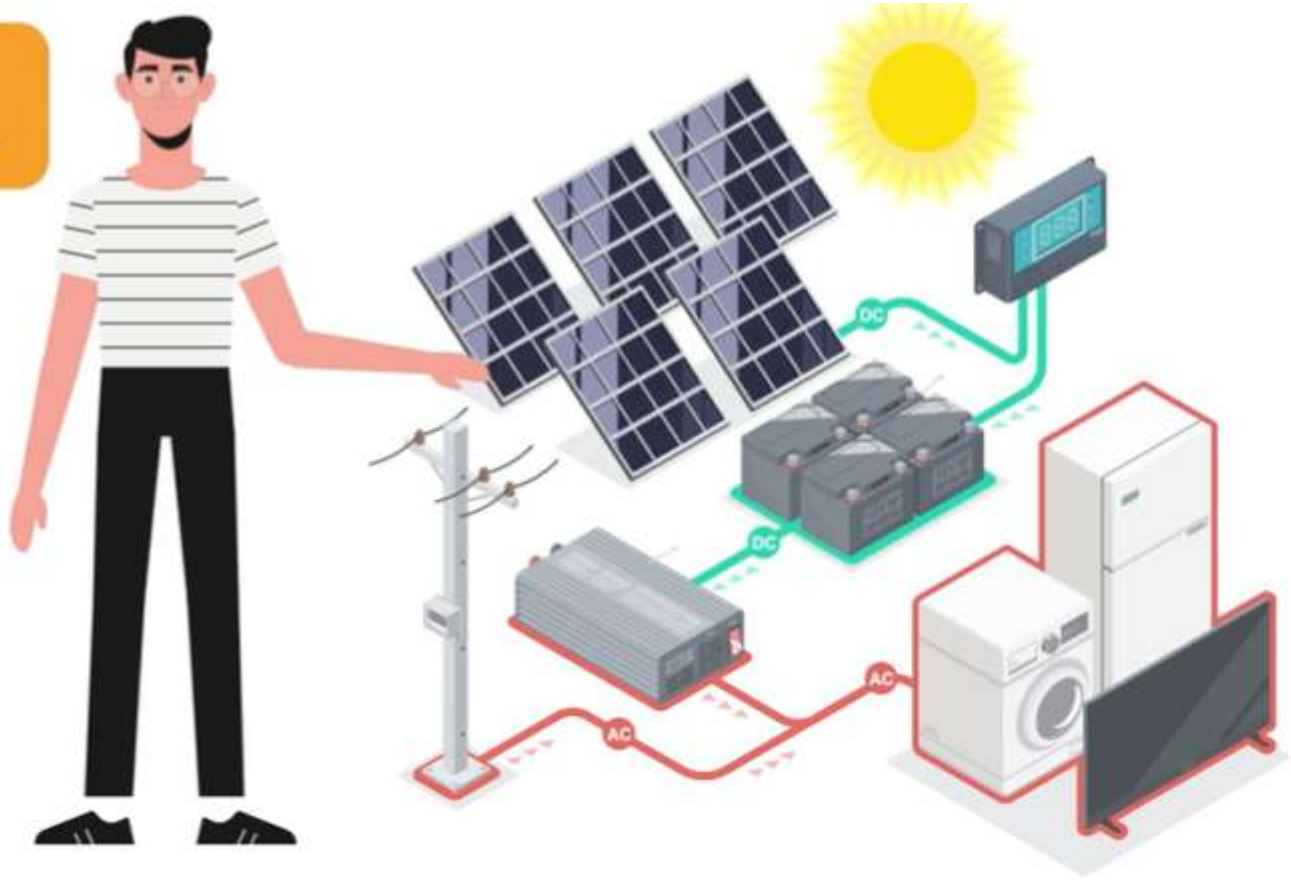
Backup Duration

Backup hours depend on availability of sunshine.





Batteries provide 6-8 hours of backup power on a sunny day.



Savings


The amount of savings you see each month will vary based on the seasons and weather.

Is there a Problem?





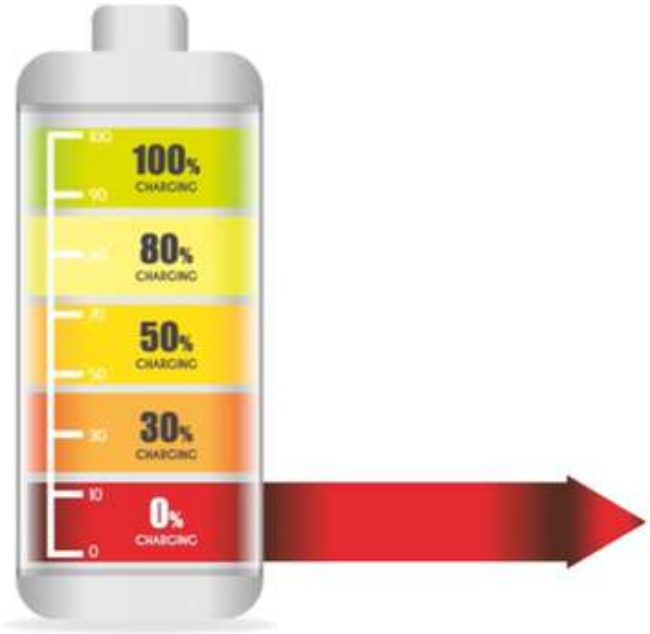


Always check your warranty details.



Benefits





Energy used daily = (Ampere-hour rating of the battery bank) × (Voltage of the bank) × (Depth of Discharge

Solar Panel Cleaning



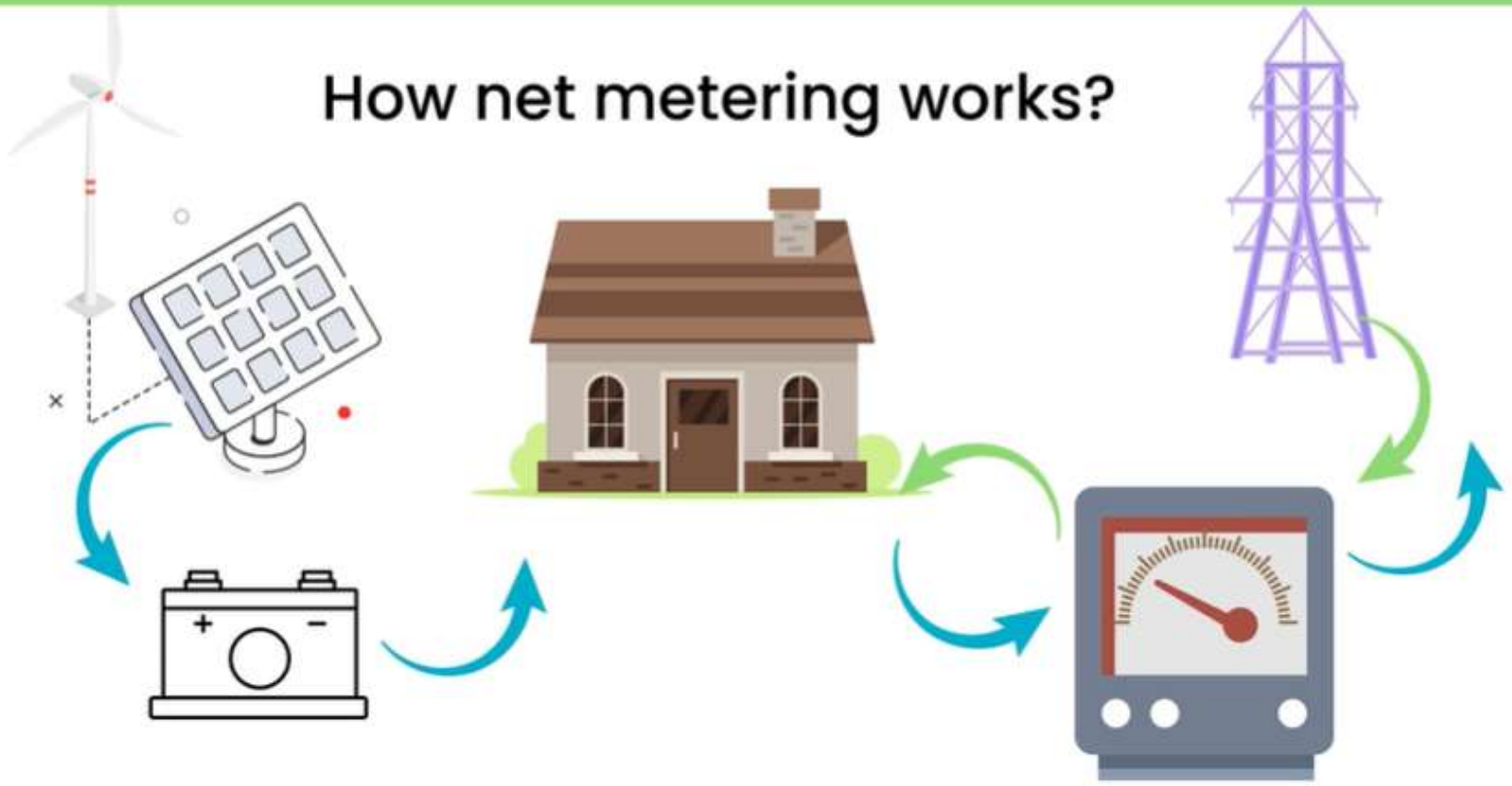
Sunrise



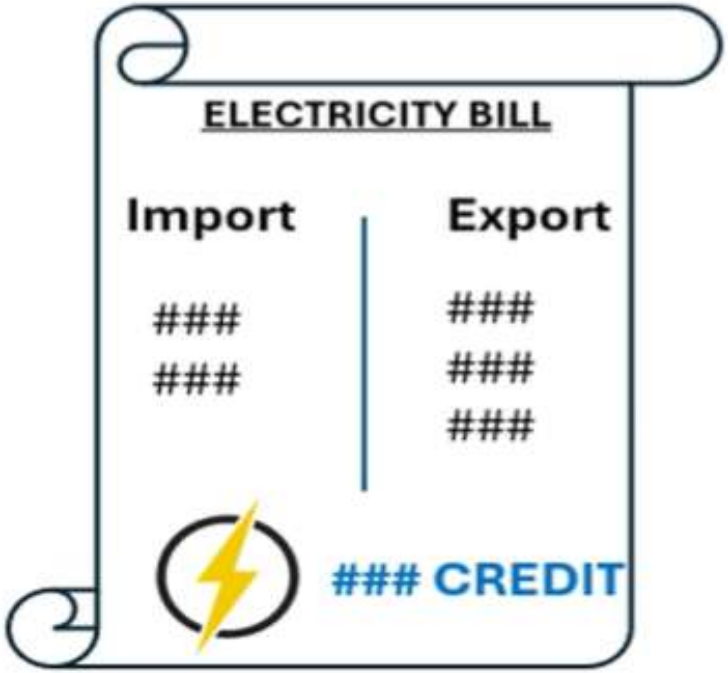
Late Afternoon



Electricity Bill and Net Metering



Reduced Overall Electricity Bill







Unit 9.1: Handover documentation on the use of the system

Hand over Documentation on the use of the Solar PV System

Includes comprehensive guidelines and instructions for the proper operation and maintenance of the system.



It covers essential aspects such as:



System Components



Safety Precautions



Troubleshooting Procedures



Contact Information for Technical Support



This documentation ensures that users understand how to maximize the efficiency and lifespan of their solar PV system while ensuring safe operation.

Handing Over Documentation





Handing Over Documentation after Solar PV Installation

Approvals and Permits



Approvals and Permits

They ensure compliance with legal and safety standards, covering

- subsidies,



- grid connection, and



- structural integrity.





They include documents such as:

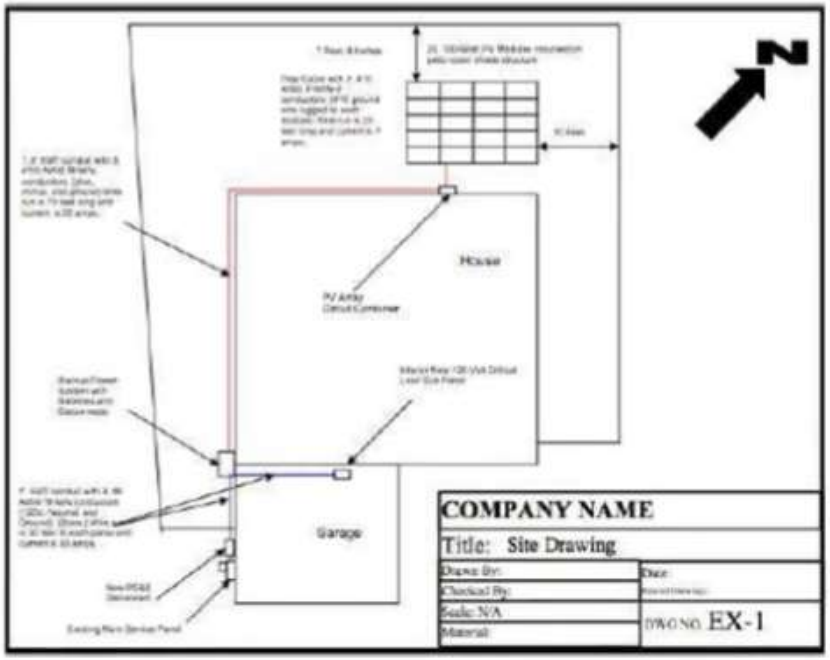
- Approval from a certified chartered engineer for building structural integrity.
- Subsidy application and approval documents, if received.
- Approval from the local utility for grid connection, must be kept ready for renewal as needed.





Layout Diagram

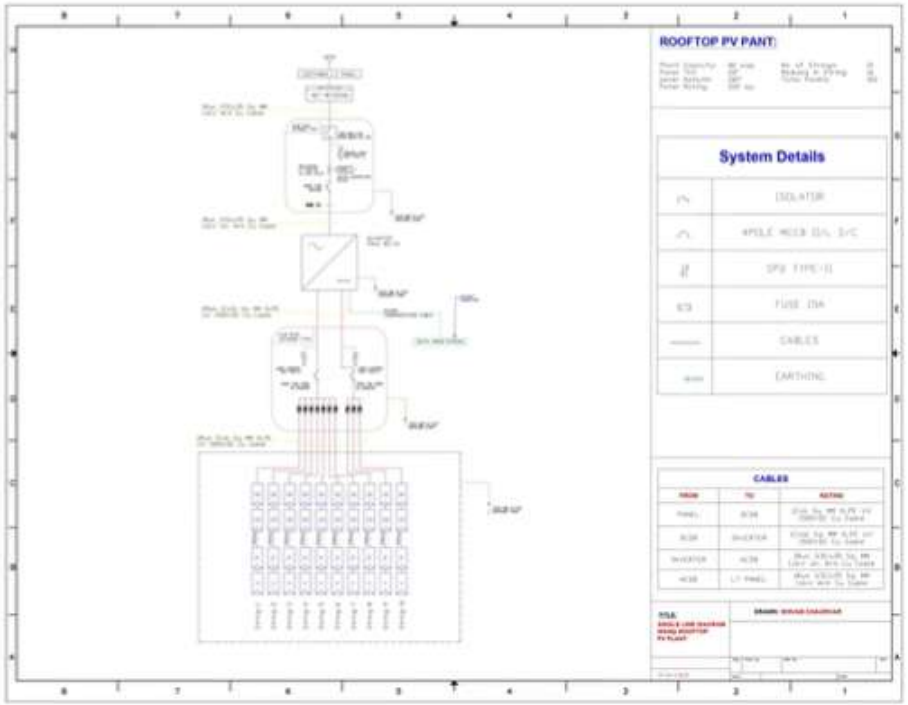
It serves as an essential visual aid and shows where different components are located for the customer's specific installation.



Single-Line Diagram

It shows the design of the system.

Both layout and single-line diagrams are useful when an engineer or technician arrives for general maintenance or troubleshooting.



Installation Certificate

Certifies that the solar panels are installed according to industry standards and regulations.



Includes important details such as the installation date and the panels' serial numbers.





Operation and Maintenance Manual

It contains instructions on how to operate and maintain your solar PV system efficiently.



It includes information on regular maintenance checks, troubleshooting, and safety precautions.





Product Documentation

It includes invoices, ratings, data sheets, and warranties for all purchased products.

They provide comprehensive information about the products used in your solar installation.

Invoices detail your purchases, while ratings and data sheets offer insights into their performance.



Warranty Certificate

It outlines the warranty coverage for your solar panels, inverters, and other components.



It's crucial to keep this safe in case you need to make any warranty claims in the future

Claim Warranty



Service contract

The service contract outlines scheduled maintenance procedures, response times for service outages, and the covered procedures.



It specifies maintenance intervals and details outage response times.



Additionally, it clarifies which system problems may incur additional costs for you as the customer.



Net Metering Agreement

It outlines the terms and conditions for connecting your solar PV system to the grid.

It helps to understand your rights and responsibilities as a solar energy producer.

